HELCOM SPIA (QGIS Plugin) - User Guide

This guide explains how to **download, install, and use** the HELCOM SPIA QGIS plugin.

Getting the Plugin

1. Open the plugin repository:

   https://github.com/yourname/ecp-raster-analysis-tool.

2. Click **Code → Download ZIP**.

3. Save the `.zip` file locally.

Installing the Plugin in QGIS

1. Open QGIS.

2. Go to: Plugins → Manage and Install Plugins → Install from ZIP.

3. Select the downloaded `.zip` file.

4. Click **Install Plugin**.

5. Enable the plugin if not enabled automatically.

After installation you should see the plugin:

- in the QGIS Plugins menu.

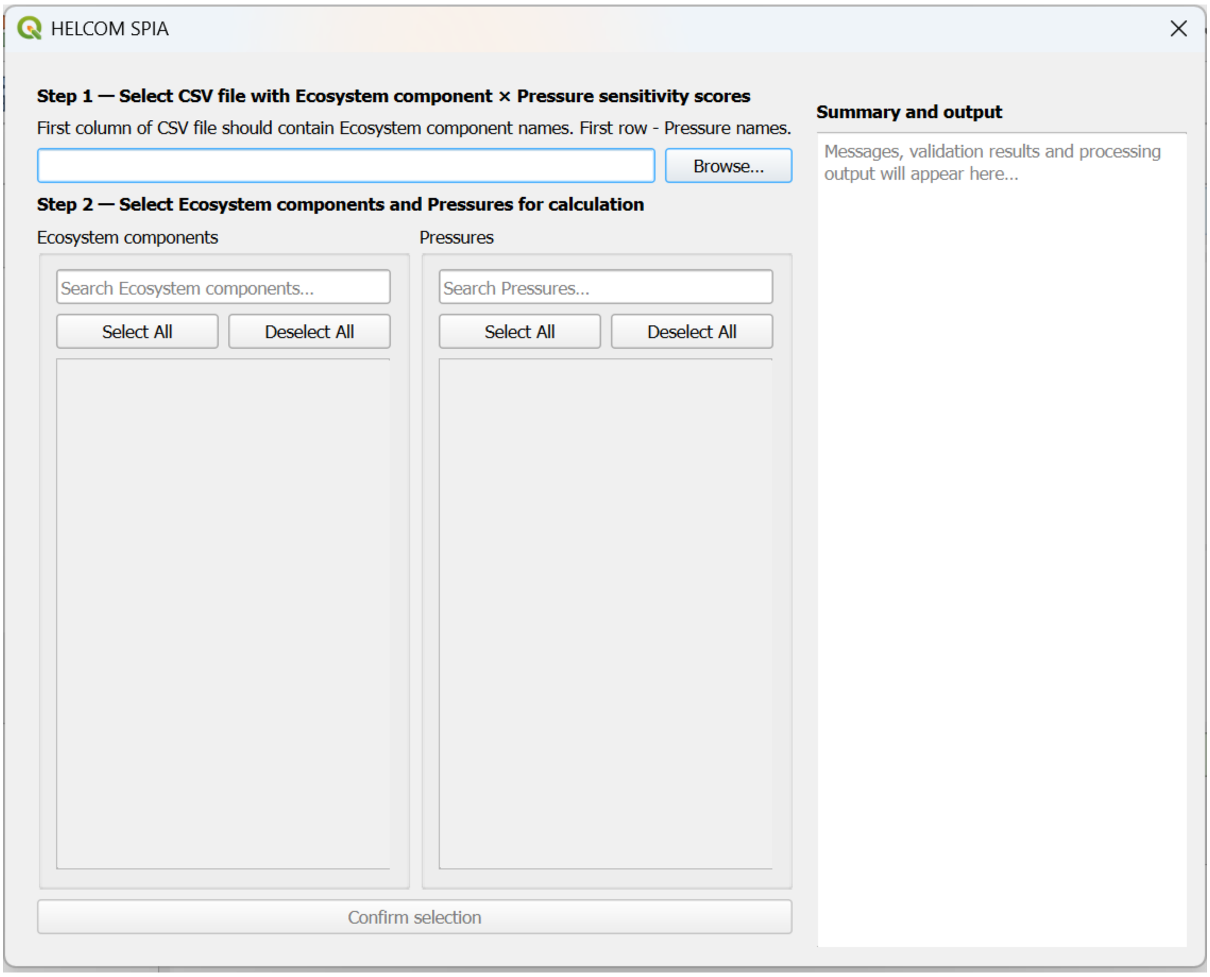
- optionally in the QGIS toolbar.

Starting the Plugin

1. Open QGIS.

2. Go to: Plugins → HELCOM SPIA.

3. The plugin dialog will open.



Overview of Workflow

The plugin guides you through a step-by-step process:

1. Load EC×P score matrix (CSV).

2. Select Ecosystem components (EC) and Pressures (P).

3. Validate raster inputs.

4. Choose processing tools.

5. Run analysis.

6. Review outputs.

Step-by-Step Usage

<u>Step 1 – Load sensitivity score matrix (CSV):</u>

- Click **Browse...**

- Select a CSV file containing EC×P sensitivity scores.

CSV structure:

- First row contains Pressure names.

- First column contains Ecosystem component names.

- Values: EC×P sensitivity scores.

Example:

	Pressure 1	Pressure 2
Ecosystem component 1	0.7	1.5
Ecosystem component 2	0	2

<u>Step 2 – Select ECs and Ps</u>

Use checkboxes to select:

- Ecosystem components.

- Pressures.

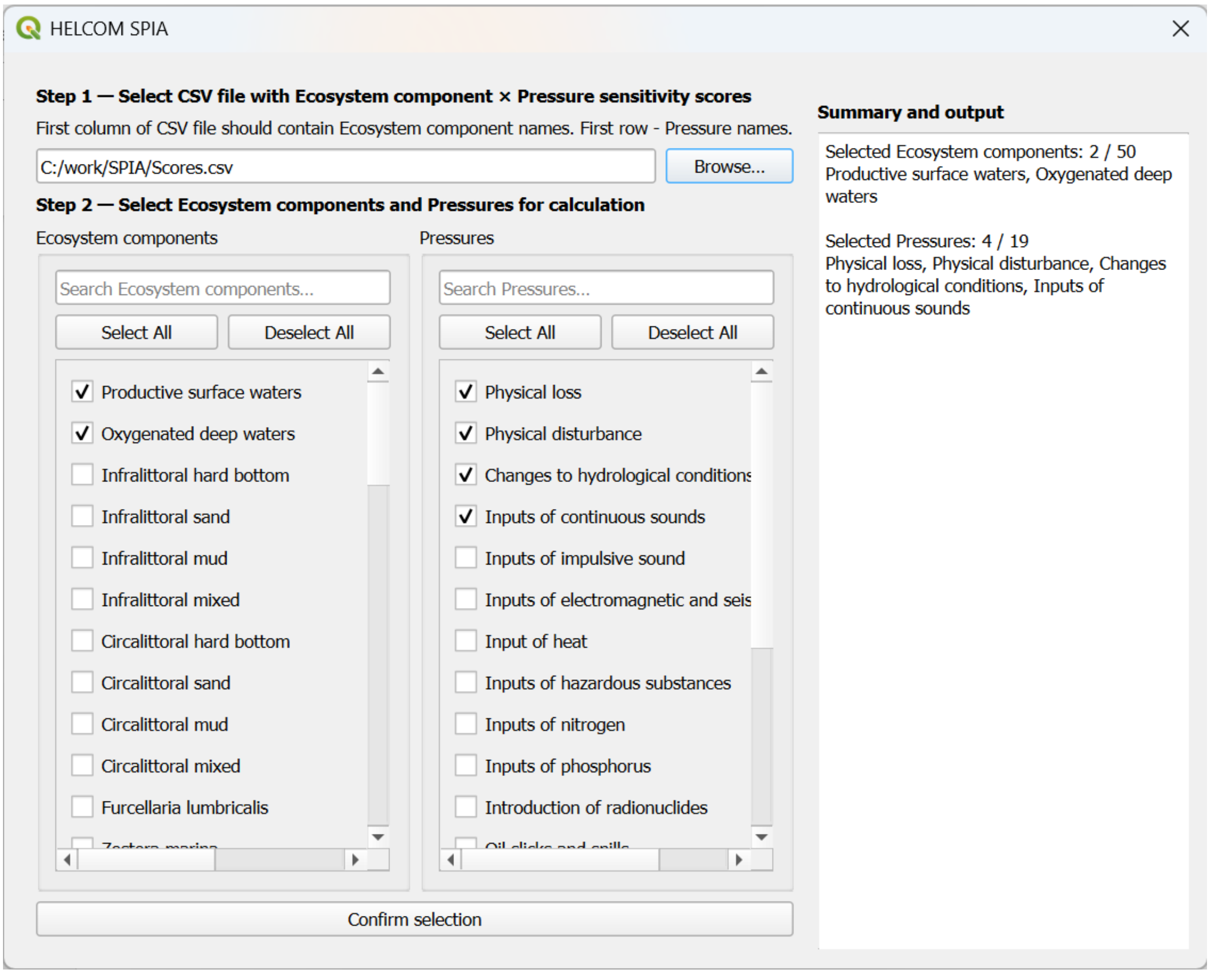
Optional:

- Use search fields to filter items.

- Use **Select All / Deselect All**.

When at least one EC and one P is selected:

- **Confirm selection** button becomes available.



Step 3 – Confirm selection

Click: **Confirm selection.**

Then:

- Selection summary appears on the right.

- Next step is unlocked.

Step 4 – Select raster folders

You must choose:

- Folder with EC rasters.

- Folder with P raster.

Each folder should contain ".tif" files **matching selected labels**.

Example:

Productive surface waters.tif

Oxygenated deep waters.tif

Physical loss.tif

Physical disturbance.tif

Step 5 – Validate raster inputs

Click: **Validate rasters.**

The plugin will check:

- File availability
- CRS consistency
- Resolution
- Dimensions
- Extent
- Data type

Possible validation outcomes:

- All good. **Possible to proceed.**
- Some selected EC and/or P rasters are missing in chosen folders. **Possible to proceed with existing.**
- No selected EC and/or P rasters in chosen folders. **Processing stopped.**
- Raster validation passed with warnings. **Possible to proceed.**
- Critical validation errors. **Processing stopped.**

Step 6 – Choose tools and output folder

Select folder where outputs will be saved.

Choose one or more SPIA tools to run:

- Impact index SUM tool

- Impact index AVERAGE tool

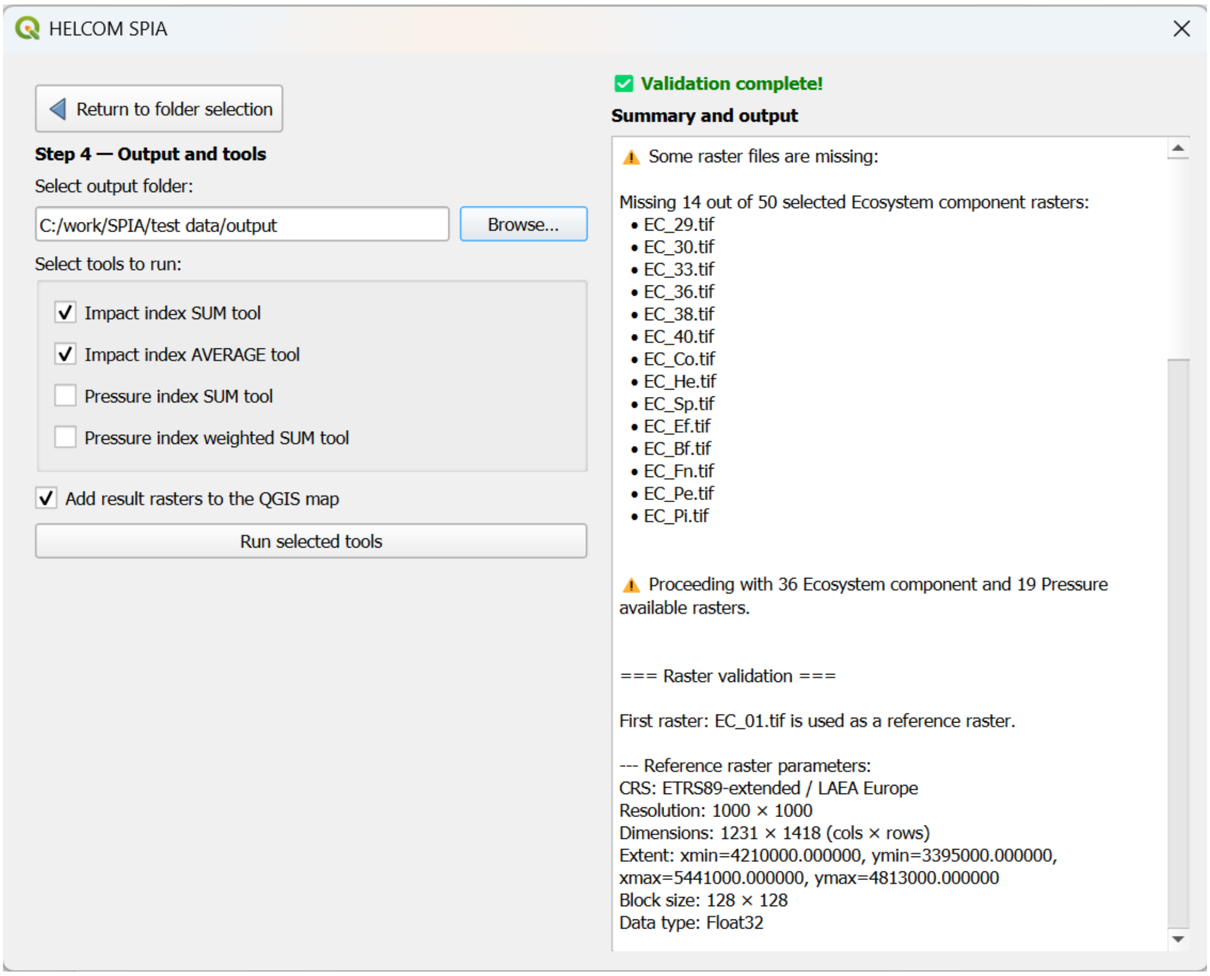
- Pressure index SUM tool

- Pressure index weighted SUM tool

The plugin will create a timestamped result folder: SPIA-results-YYYY-MM-DD-HH-MM-SS in the selected output folder. Tools produce following outputs:

Tool	Output TIF file	Output CSV
Impact index SUM tool	Impact-index-SUM-YYYY-MM-DD-HH-MM-SS.tif	Contribution-matrix-YYYY-MM-DD-HH-MM-SS.csv
Impact index AVERAGE tool	Impact-index-AVERAGE-YYYY-MM-DD-HH-MM-SS.tif	Contribution-matrix-YYYY-MM-DD-HH-MM-SS.csv
Pressure index SUM tool	Pressure-index-SUM-YYYY-MM-DD-HH-MM-SS.tif	
Pressure index weighted SUM tool	Pressure-index-weighted-SUM-YYYY-MM-DD-HH-MM-SS.tif	

If **Add result rasters to the QGIS map** checkbox is checked, produced TIF file can be added to QGIS map.



Step 7 – Run processing

Click **Run selected tools.**

During processing progress bar shows progress and estimated processing time.

To cancel processing click **Cancel processing.** Processing will stop safely and no incomplete outputs will be written.

After processing is finished:

- Generated output files are listed.

- Raster layers are added to the map (if **Add result rasters to the QGIS map** checkbox is checked).

- Button **Start the whole processing again** appears. Processing can be repeated with different input files.

- Plugin can be closed by clicking on the X in the top right.

Performance Tips

- Impact index SUM tool and Impact index AVERAGE tool share computation. It is faster to run them together.

- Large datasets may take many hours. Consider running tool with many large datasets during the night or weekend.

Common Issues

No rasters found:

- Check folder paths

- Check file names match CSV labels

Validation fails:

- Ensure rasters share CRS, resolution, extent

Processing is slow:

- Reduce number of ECs / Ps

- Use smaller raster datasets for testing

Best Practices

- Start with small datasets

- Validate inputs before running

- Use timestamped outputs to compare runs

- Keep EC/P naming consistent